KUSH MEHTA

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Final-year Electronics & Communication Engineering student with hands-on C/C++ systems experience from a research internship at ISRO — real-time protocol decoding, error-correcting codes, and TCP/IP streaming under hard toolchain constraints — plus three years leading embedded avionics and firmware for competition UAVs.

EDUCATION

Institute of Technology, Nirma University Ahmedabad, India
B.Tech, Electronics & Communication Engineering — Minor: Data Science | CGPA: 8.59 / 10 Aug 2023 – May 2027

- **Coursework:** Embedded Systems, Computer Architecture, Microcontrollers & Interfacing, Scripting Languages, VLSI Design, Digital Signal Processing, Programming for Scientific Computing

EXPERIENCE

Research Intern — Navigation Receiver Division, Space Applications Centre (SAC), ISRO May – Jul 2026
Real-Time Galileo High Accuracy Service (HAS) Decoding Utility · 10-week research internship Ahmedabad, India

- Built a real-time C++ utility that ingests a multiplexed TCP stream, tokenizes three interleaved protocols (RTCM payloads, Galileo HAS E6-B pages, Time-of-Transfer epoch tags), and routes each to independent output ports.
- Implemented Reed–Solomon (255, 32) erasure decoding over GF(2^8) from the Galileo HAS ICD — lookup-table field arithmetic and Gaussian elimination for k×k matrix inversion — recovering complete messages despite page loss and out-of-order arrival.
- Reverse-engineered four undocumented Galileo RTCM 3.x message formats (1240–1243) and the CRC-24Q framing routine by cross-referencing an open-source implementation against the ICD.
- Delivered streaming over WinSock 2.2 TCP and Win32 serial with background-thread reads and stale-buffer purging, working around a pre-C++11 toolchain that ruled out modern STL containers and threading.
- Validated end-to-end in RTKLIB/RTKNAVI at centimetre-level Precise Point Positioning, with each decoding stage unit-verified against a Python reference implementation.

Electronics Lead — Team Arrow UAV Club, Nirma University Sep 2023 – Present
Official Aeromodelling & Autonomous Systems Club Ahmedabad, India

- Own the avionics and embedded stack for competition aircraft: flight-controller configuration (ArduPilot / Pixhawk), telemetry links, sensor integration, ESC and motor selection, and power system design.
- Designed autonomous payload electronics — magnetometer, time-of-flight ranging, IR line-following — for a ground vehicle deployed from a tilt-rotor VTOL aircraft; validated the mission in software-in-the-loop before hardware.
- Lead a team of 8–12 and mentor juniors on firmware, sensor fusion, and avionics integration. Four podium finishes across national and international SAE and Technoxian competitions, including AIR 2 (SAE DDC Micro 2024) and 5th internationally in mission score (SAE Aero Design West 2025, USA).

PROJECTS

Secure GPS Telemetry Pipeline — ESP32, AES-128, MQTT Mar 2026

- End-to-end encrypted IoT pipeline: ESP32 parses NMEA over UART2 with TinyGPS++, serializes to JSON, encrypts with AES-128 via mbedTLS, and publishes hex ciphertext to an MQTT broker on a 3-second cycle.
- Python subscriber (paho-mqtt, PyCryptodome) decrypts and resolves the live position; verified on hardware from GPS fix through encrypted transport to decoded coordinates.

gem5 L1 Cache Sensitivity Analysis Feb 2026

- Three-experiment microarchitectural study on gem5 v25.1 (X86 TimingSimpleCPU) varying cache capacity, workload type, and set associativity.
- Cache growth 4 kB→64 kB cut matrix-multiply miss rate 5.06%→0.04% (IPC +11.9%); 1-way→8-way associativity cut it 4.74%→0.34% (IPC +10.4%), while an AES workload stayed flat at 0.14% — isolating execution units, not memory, as its bottleneck.

Cryptography Benchmark for Resource-Constrained Devices Jan 2026

- Benchmarked AES-GCM, DES, ECC P-256, and ASCON-128 across 5,000 IoT payloads; loaded the ASCON C reference as a compiled DLL through Python `ctypes` for native-speed timing, profiling peak memory with `tracemalloc`. ASCON ran 78× faster than AES-GCM and 12× faster than DES.

Log Analytics & Anomaly Detection Tool — Bash, Python, Linux Mar 2026

- Two-process monitoring tool: a Bash collector samples CPU, memory, disk, and failed-login metrics on a cron schedule; a Python detector applies a $\mu+2\sigma$ baseline threshold and dispatches alerts on breach. Validated under synthetic CPU stress, with the cold-start threshold limitation documented.

TECHNICAL SKILLS

Languages: C, C++, Python, Verilog HDL, Bash, SQL, Assembly, MATLAB, TCL

Embedded & Hardware: ESP32, Raspberry Pi, Arduino, 8051, Pixhawk flight controllers, ArduPilot / Mission Planner, Keil uVision, mbedTLS, UART / I²C / SPI, FPGA

Systems & Networking: Linux / Ubuntu, WinSock TCP/IP, Win32 API, multithreading, cron, MQTT (HiveMQ, Mosquitto), Wi-Fi, BLE

Standards & Algorithms: Reed–Solomon, Galois Field GF(256), Galileo HAS ICD, RTCM 3.x, CRC-24Q, AES-128, ASCON-128, ECC P-256

Tools: Git / GitHub, gem5, ModelSim, Quartus II, RTKLIB, pandas, NumPy, OpenCV, scikit-learn, PyTorch